Abdulkadir Elaldı

Embedded Systems · AI · Full-Stack

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Summary

I build end-to-end systems where hardware meets software: from STM32-based control software to OCR systems running in the field and AI models designed to run on microcontrollers.

Awards and achievements

1st place in Türkiye — TEKNOFEST 2026 Hyperloop Development Competition · Technology Demonstration (Spectraloop)

3rd place in Türkiye — TEKNOFEST 2026 Hyperloop Development Competition · Performance (Spectraloop)

3rd place in Türkiye — TEKNOFEST 2026 Hyperloop Development Competition · Defined Problem Solving (Spectraloop)

1st place, Technical Design Report — TEKNOFEST 2026 Hyperloop · Performance Category Technical Design Report (Spectraloop · 468.5/500)

Mechanical System Development Award — TEKNOFEST 2025 Hyperloop Development Competition · Finalist (Spectraloop)

1st place in Samsun — TUA Astro Hackathon · CubeSat (Team Captain)

Finalist — TEKNOFEST 2021 Unmanned Underwater Systems Competition (Provezza · Software Team Lead)

1st place — Deneyap Technology Festival · Robotic Coding, Erzurum (Team Captain)

Experience

Co-Founder & Head of Software · Kıyı Medya — 360° digital agency

Sep 2024 — Present

I was part of founding the agency and run its software unit; the agency has delivered more than 500 projects to more than 50 companies. I built the agency's corporate websites end to end (İSÇEV Arıtma, Aşkar Medikal, Costman, İlkadım Hearing Aids, CRP Project and others) and am leading the development of a mobile app.

Software Architect & Team Lead · Spectraloop Technology Team — TEKNOFEST Hyperloop

Nov 2023 — Present

I have led the software team for 3 seasons, coordinating the subsystem groups (sensors, CAN, navigation, BMS, real-time control, telemetry) and the Technical Design Report process; in 2026 the report placed 1st with a score of 468.5 out of 500. I developed the STM32 firmware that drives two Delta MS300 VFDs over Modbus RTU on a single RS-485 line; the propulsion and braking systems were field-tested at the TÜBİTAK Gebze campus.

Software Team Lead · SEA STARS Team — TEKNOFEST Unmanned Underwater Systems

Sep 2024 — Sep 2026

I led the team developing the mission software for an autonomous underwater vehicle: serial communication between Jetson Nano, STM32, and Arduino, plus YOLO / TensorFlow-based object detection. The TEKNOFEST 2026 Critical Design Report stage was completed with a score of 83.1.

Intern · Adapha Yapay Zeka Ar-Ge ve Yazılım A.Ş. (Samsun Teknopark)

Jul 2026 — Aug 2026

Two end-to-end projects: (1) JWC Line Monitoring System — a Raspberry Pi 5 field device and a GStreamer video pipeline, with RapidOCR / ONNX Runtime on ARM reading all 6 fields in the field at 99%+ confidence; a FastAPI event engine and real-time OEE pushed over WebSocket to React Native dashboards. (2) Radar — a cloud-based lead screening and audit system that analyzes and reports on business accounts: async FastAPI, PostgreSQL, Redis, Playwright, and Next.js, with deterministic rules backed by a two-stage LLM (Anthropic).

Founder & CEO · TesisBul.com

Jun 2024 — Feb 2025

I built from scratch a platform for booking sports facilities such as artificial-turf football pitches online, with or without prepayment, and was responsible for how the product operated and its strategic development. I ran marketing and field outreach targeting more than 100 businesses in about 20 cities.

Software Team Lead · Provezza Team — TEKNOFEST Unmanned Underwater Systems

Jan 2021 — Dec 2021

In high school, I led the software team of an autonomous underwater vehicle; using Jetson Nano, Arduino, STM32, and YOLO / TensorFlow, we became TEKNOFEST finalists.

Education

B.S. in Software Engineering — 4th year · Samsun University

Oct 2023 — Present

Expected graduation: June 2028.

Faculty of Law (Undergraduate) · Atatürk University

Jan 2022 — Dec 2023

Transferred to Software Engineering at Samsun University under the Additional Article 1 transfer provision.

Graduate · Deneyap Technology Workshops

Jan 2019 — Dec 2022

Robotic coding, mobile apps, nanotechnology and materials science, and design and manufacturing.

Certifications

Aviation Engines School (6th Cohort) — TEI — TUSAŞ Engine Industries (Jun 2026)

ASP.NET Core Web API Training — BTK Akademi (Jan 2025)

Introduction to Web Services Training — BTK Akademi (Nov 2024)

Entrepreneurship Training — KOSGEB (Nov 2024)

Mobile App Development Training — Deneyap Türkiye (May 2022)

Unmanned Underwater Systems Competition Participation Certificate — TEKNOFEST (Sep 2021)

Design and Manufacturing Training — Deneyap Türkiye (Sep 2019)

TEKNOFEST Hyperloop Development Competition Finalist — TEKNOFEST (Sep 2025)

ASP.NET Core MVC Training — BTK Akademi (Jan 2025)

Object-Oriented Programming with Applications Training — BTK Akademi (Nov 2024)

Volunteer Staff Certificate of Appreciation — TEKNOFEST (Apr 2023)

Nanotechnology and Materials Science Training — Deneyap Türkiye (Feb 2022)

Robotic Coding Training — Deneyap Türkiye (Dec 2019)

Sports

2013 Champion — Erzurum Inter-Primary School Sports Festival

2014 Medal — Erzurum

2015 Champion — Yıldızkent, youth category

2016 3rd place — Inter-Middle School Tournament

2018 Champion — Turkish Tennis Federation · Batman Eastern League

2019 2nd place — Inter-High School Tournament

2024 1st place — Samsun University · 2023–2024 Spring Semester Tournament

2024 1st place — Samsun University · 2024–2025 Fall Semester Tournament

2025 3rd place (Captain) — Samsun University · 2024–2025 Spring Semester Tournament

2025 2nd place — Samsun University · 2024–2025 Spring Semester Tournament

2025 1st place — Samsun University · 2025–2026 Fall Semester Tournament

2026 1st place — Samsun University · 2025–2026 Spring Semester Tournament

Skills

Languages C · C++ · Python · TypeScript · JavaScript · C#

Embedded systems STM32 (HAL) · Raspberry Pi 5 · Jetson Nano · Arduino · CAN Bus · Modbus RTU / RS-485 · UART · I²C · Interrupts & Watchdog

AI / computer vision OCR (RapidOCR / PaddleOCR) · OpenCV · YOLO · TensorFlow · scikit-learn · ONNX Runtime · TinyML · LLM integration (Gemini, Anthropic)

Frontend React · Next.js · HTML / CSS

Backend	FastAPI · Flask · Node.js · Express · tRPC · Prisma · SQLAlchemy · ASP.NET Core
Databases	PostgreSQL · MongoDB · SQLite · Redis
Mobile	Mobile app development · React Native
Tools	Git / GitHub · Linux · systemd · GStreamer · WebSocket · Playwright · Claude Code